



OPEN Prediction of recurrence after resection in hepatocellular carcinoma via whole liver deep learning on preoperative contrast-enhanced CT

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This study aimed to develop a fully automated survival prediction (FASP) system that analyzes whole-liver regions from preoperative contrast-enhanced CT scans for predicting recurrence-free survival (RFS) after curative resection in Hepatocellular carcinoma (HCC). FASP comprised three consecutive components: automatic liver and tumor segmentation models, and a RFS prediction model, LiverSurv, all based on deep convolutional neural networks. FASP was compared against a clinical model leveraging clinical factors and three tumor-based methods using semantic, radiomic, or deep learning features. A total of 827 patients were included across the development, internal test, and external test sets. In the internal and external test sets, FASP achieved concordance indices (C-indices) of 0.646 (95% CI: 0.566, 0.725) and 0.786 (95% CI: 0.726, 0.846), respectively, outperforming the clinical model (both adjusted $P < .05$). Integrating clinical factors improved C-indices to 0.664 (95% CI: 0.591, 0.736) internally and 0.800 (95% CI: 0.750, 0.849) externally. FASP also surpassed the tumor-based models, which yielded C-indices ranging from 0.623 to 0.632 internally and 0.523 to 0.775 externally. Visualization analysis demonstrated that FASP captured prognostic information from both tumor and background liver regions. These findings suggest whole-liver-based deep learning provides a promising non-invasive approach to predict recurrence risk for HCC patients before surgery.

Hepatocellular carcinoma (HCC) accounts for 75%–85% of primary liver cancer, which is the third leading cause of cancer deaths worldwide¹. Curative resection is a first-line therapy for HCC yet the 5-year recurrence rate approaches 70%². Accurate preoperative prediction of recurrence risk is increasingly critical for optimizing treatment planning². However, no universally accepted prognostic system exists for predicting HCC recurrence before resection³. Conventional staging systems offer limited prognostic power because they rely on expert consensus rather than robust statistical methodologies³. Several clinical models have been developed using parameters from routine medical records and rigorous statistical approaches^{3–5}. Advances in medical imaging and artificial intelligence (AI) have enabled more accurate preoperative prediction of recurrence in HCC^{6–8}. However, most imaging-based studies focus on the largest tumors while neglecting the critical role of other tumors and background liver conditions in recurrence.

Multifactorial evidence emphasizes the need for a comprehensive whole-liver assessment in HCC. First, multifocal tumors are found in 41%–75% of HCC patients at diagnosis, indicating a widespread disease burden within the liver⁹. Second, pathological features at tumor margins, such as microvascular invasion, significantly

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increase the likelihood of tumor spread¹⁰. Third, the residual liver often harbors disseminated tumor cells or undetectable tumors that serve as seeds for tumor relapse^{9,10}. Furthermore, most HCC patients have chronic liver diseases, particularly cirrhosis, which creates a persistently inflamed and tumor-promoting environment^{11,12}. Collectively, these factors highlight the elevated risk of tumor recurrence throughout the liver.

Quantitative imaging techniques for predicting HCC recurrence have primarily adopted tumor-focused strategies using radiomics or deep learning methods^{6–8,13}. Radiomics-based approaches typically involve extracting a large number of hand-crafted tumor features, selecting relevant features, and fitting them with traditional machine learning algorithms^{6,13}. In contrast, deep learning methods automatically learned intricate features from tumor regions to predict recurrence risk^{7,8}. While these studies have advanced HCC recurrence prediction, they often focus solely on the largest tumor, neglecting smaller tumors and the background liver parenchyma.

In this study, we developed and validated a fully automated survival prediction (FASP) system to mine whole-liver information from preoperative contrast-enhanced computed tomography (CECT) for robust predictions of recurrence-free survival (RFS) in HCC patients undergoing curative resection. Furthermore, we investigated the model interpretability of this whole-liver approach, offering key insights into its clinical relevance for HCC recurrence prediction.

Materials and methods

Study design and participants

This retrospective study was approved by the Clinical Research Ethics Committee of the First Affiliated Hospital of Zhejiang University (FAHZU), which also granted a waiver of written informed consent. It was conducted in accordance with the Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD) guidelines¹⁴.

The overall study design is shown in Fig. 1. This study comprised participants from three sources: (1) FAHZU: HCC patients who underwent radical (R0) resection between June 2014 and June 2022. R0 resection was defined as complete tumor removal with no residual tumor detected during surgery and negative margins on pathological examination of frozen sections. All patients were followed up until April 27, 2023. (2) Huzhou Central Hospital: HCC patients who underwent R0 resection between April 2014 and January 2020. All patients were followed up until October 20, 2022. Both FAHZU and Huzhou Central Hospital are tertiary-care hospitals. (3) The Cancer Genome Atlas Liver Hepatocellular Carcinoma (TCGA-LIHC) data collection which is publicly available.

Supplementary Fig. S1 illustrates the flowchart of patient inclusion and exclusion, as well as dataset partition for this study. More details are presented in Supplementary Methods. Patients from FAHZU were randomly split into development and internal test sets at a ratio of 4:1. To address sample imbalance, stratified sampling was performed based on year of surgery, CECT slice spacing, and postoperative trans-arterial chemoembolization (TACE) to ensure balanced event proportions across subsets. Due to small sample sizes, the other two datasets were combined into a single external test set.

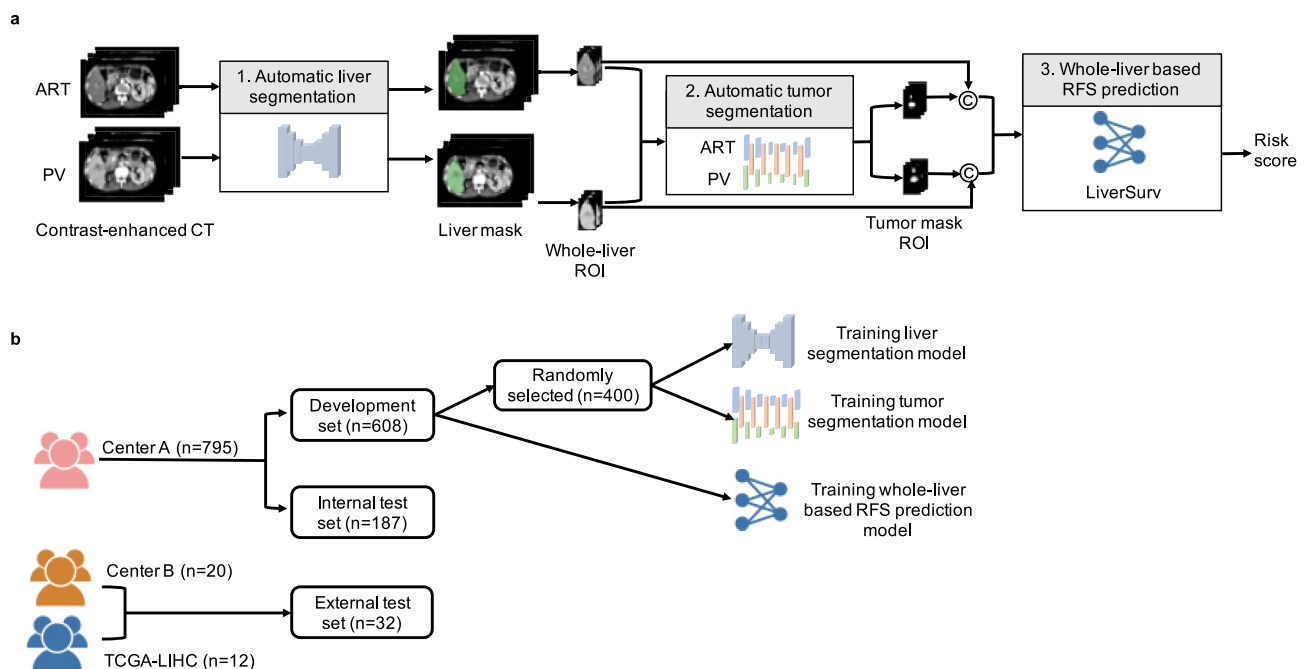


Fig. 1. Workflow of the proposed FASP and study design. **(a)** Schematic workflow of the whole-liver based FASP in predicting RFS in HCC patients treated with curative resection. **(b)** Datasets used in this work.

The primary endpoint was recurrence, diagnosed according to established criteria¹⁵. RFS was defined as the time (in months) from the date of surgical resection to the first detected tumor recurrence, with patients without recurrence censored at death unrelated to HCC or at last follow-up.

CECT scanning protocols and image preprocessing

The acquisition parameters for CECT are summarized in Supplementary Table S1. The arterial (ART) and portal venous (PV) phases of preoperative CECT scans were extracted for analysis. Preprocessing steps for the images are detailed in the Supplementary Methods.

Development of FASP and FASP-clin

The workflow of FASP is illustrated in Fig. 1. FASP consisted of three consecutive components: an automatic liver segmentation model, an automatic tumor segmentation model, and a RFS prediction model named LiverSurv. All components were based on deep convolutional neural networks (CNNs). Using the ART and PV phases of preoperative CECT scans as input, the liver and tumor segmentation models first generated two types of regions of interest (ROIs): whole liver ROIs, where liver voxels retained their original CT intensities while background voxels were set to 0, and tumor mask ROIs, in which voxels were 1 for tumor and 0 for background. For each phase, the whole liver and tumor mask ROIs were concatenated, and the combined ROIs from both phases were simultaneously forwarded to the LiverSurv model to produce a patient-specific recurrence risk score. Further details on the development and architecture of these components are provided below.

The LiverSurv model employed two parallel CNN-based branches to integrate features from both ART and PV phases for robust RFS prediction (Supplementary Fig. S4). For each patient, whole liver and tumor mask ROIs from each phase were concatenated and fed into the corresponding phase-specific branch, where high-dimensional image features were extracted via three-dimensional (3D) convolution blocks. The resulting feature maps were flattened, concatenated, and processed through five fully connected layers. Two specialized modules were incorporated to enhance feature extraction: inter-phase attention (IPA) and multi-scale feature extraction (MSFE) (Supplementary Fig. S4). IPA, inspired by the Squeeze-and-Excitation networks¹⁶ and Multimodal transfer module¹⁷, facilitated information exchange between phases. MSFE used a couple of scale-specific feature extractors (SSFE) with global max-pooling, convolution and fully connected layers to capture recurrence-relevant features at multiple resolutions, mitigating information loss from repeated downsampling. The output of LiverSurv was a “risk score” representing the logarithm of the hazard ratio relative to baseline hazards. Unlike conventional Cox proportional-hazards models, which rely on linear functions, LiverSurv employed a highly non-linear function to analyze unstructured imaging data. Cox proportional-hazards loss was used to train LiverSurv¹¹.

Additional details on whole-liver ROI preparation, LiverSurv training, as well as the development of the liver and tumor segmentation models are provided in Supplementary Methods and Fig. S2. Segmentation performance was evaluated using Dice coefficient, Intersection-Over-Union (IoU), Hausdorff distance (HD95), and average surface distance (ASD).

Additionally, FASP-Clin was developed by integrating the FASP score with established clinical prognostic indicators of RFS in HCC, including AFP, ALBI score, and cirrhosis status. These factors were selected for their noninvasive nature, preoperative availability, and the fact that they do not require cumbersome manual assessments. A separate clinical model incorporating these clinical factors was also developed for comparison. All models were built using multivariable Cox proportional hazards regression, as detailed in the Statistical Analysis section.

Comparison of FASP with tumor-based models

For comparison with tumor-based methods, we developed three tumor-based models using conventional strategies: tumorSemantics, tumorRadiomics, and tumorSurv (Supplementary Fig. S5). All the models focused exclusively on the largest tumor regions which were manually annotated by professional radiologists. The tumorSemantics model used radiologist-defined semantic features, including tumor diameter and multifocality. The tumorRadiomics model utilized radiomic features extracted from the annotated tumor regions, covering intensity, shape, size, texture, and wavelet-based characteristics. The tumorSurv model was a convolutional neural network (CNN) that analyzed tumor ROIs derived from the radiologist-defined masks. Further details on developing the tumor-based models are shown in Supplementary methods.

Investigation of model interpretability

To explore how FASP generates predictions, we examined the associations between the FASP score and conventional staging systems and pathological risk factors. Additionally, we applied the gradient-weighted class activation mapping (Grad-CAM) algorithm to generate class saliency maps (CAMs) to highlight the liver regions that contributed most to FASP’s predictions. Details on Grad-CAM implementation are available in Supplementary methods and Fig. S8.

Statistical analysis

Statistical analyses were carried out using R (version 4.3.2, <https://www.R-project.org>, RRID: SCR_001905). Differences in patient characteristics across datasets were assessed using chi-square tests for categorical variables and Kruskal-Wallis tests for continuous variables. Correlations between the FASP score and pathological risk factors were evaluated with Spearman’s rank correlation.

In the multivariable Cox proportional hazards regression analysis, feature selection was performed via backward elimination based on Akaike information criteria¹⁸. Time-dependent receiver operating characteristic (ROC) curves and corresponding area under the curve (AUC) values were used to assess the dynamic performance

at different time points by using the “timeROC” (RRID:SCR_026444) package. For each survival model, patients were stratified into low-risk and high-risk groups based on the median predicted risk score of the development set. Kaplan-Meier (KM) survival curves and log-rank tests were used to evaluate differences between these groups using the “survival” (RRID:SCR_021137) and “survminer” (RRID:SCR_021094) packages¹⁹.

Survival analyses using non-deep learning methods were conducted with univariable and multivariable Cox proportional hazards regression using the “survival” package. The primary assessment criterion was Harrell’s C-index²⁰. C-index estimates the proportion of patient pairs whose predicted relative risks were correctly ranked as observed²¹, reflecting the probability that, for any two randomly selected patients, the one with a higher risk score will have a shorter survival time²⁰. A C-index of 1.0 indicates perfect prediction while a C-index near 0.50 suggests random prediction²¹. The C-index and its 95% confidence interval (CI) were computed using the “survcomp” (RRID:SCR_003054) package. Comparisons of C-indices between models were conducted using the dependent Student’s t-test based on bootstrapping, following a recent study²². A two-sided *P* value < 0.05 was considered statistically significant. Bonferroni correction was applied to adjust *P* values for multiple comparisons.

Results

Patient characteristics

The development, internal test and external test sets included 608, 187 and 32 patients, respectively. Baseline and survival characteristics of the patients were summarized in Table 1. The development and internal test sets were comparable across most characteristics, whereas the external test set exhibited distinct heterogeneity. Specifically, patients in the external test set had larger tumors, higher ALBI scores, and higher recurrence rates, with a greater proportion experiencing early recurrence which is defined as relapse within two years after resection. Moreover, the external test set had significantly larger slice spacings in their CT scans.

Performance of the automatic liver and tumor segmentation models

As shown in Supplementary Table S2 and Fig. S3, the automatic liver segmentation model achieved a mean Dice score of 98.3% across both phases. The tumor segmentation model demonstrated promising performance, achieving a mean Dice score of 72.2% across both phases (Supplementary Table S3).

Characteristic	Development set (N=608)	Internal test set (N=187)	External test set (N=32)	P Value
Age, year	59.0 (52.0, 65.0)	58.0 (51.0, 64.0)	57.0 (52.8, 65.0)	.80
Male	515 (84.7%)	155 (82.9%)	21 (65.6%)	.02
Liver cirrhosis	470 (77.3%)	149 (79.7%)	23 (71.9%)	.58
Multifocality	38 (6.2%)	14 (7.5%)	3 (9.4%)	.56
Tumor diameter ^a , cm	3.5 (2.5, 5.0)	3.5 (2.5, 5.5)	5.2 (2.9, 9.2)	.04
Tumor diameter group				
≤2	104 (17.1%)	30 (16.0%)	3 (9.4%)	.005
>2 and ≤5	357 (58.7%)	107 (57.2%)	12 (37.5%)	
>5 and ≤10	123 (20.2%)	41 (21.9%)	10 (31.2%)	
>10	24 (3.9%)	9 (4.8%)	7 (21.9%)	
Tumor volume of ART phase, cm ³	23.7 (9.6, 59.8)	23.0 (9.4, 78.6)	41.1 (12.6, 212.4)	.15
Tumor volume of PV phase, cm ³	23.5 (9.6, 57.2)	22.2 (9.5, 80.0)	39.6 (12.3, 212.0)	.15
AFP≥400 ng/ml	150 (24.7%)	39 (20.9%)	7 (21.9%)	.55
ALBI	−3.0 (−3.2, −2.7)	−3.0 (−3.2, −2.7)	−2.5 (−2.8, −0.5)	<.001
Postoperative TACE	309 (50.8%)	95 (50.8%)	NA	NA
Recurrence events	210 (34.5%)	63 (33.7%)	22 (68.8%)	<.001
Follow-up time, month	29.7 (11.9, 43.3)	30.1 (12.2, 43.3)	26.6 (5.3, 76.7)	.92
Recurrence rate				
2 year	152/467 (32.5%)	47/149 (31.5%)	13/22 (59.1%)	.03
5 year	207/259 (79.9%)	62/76 (81.6%)	21/30 (70.0%)	.39
CECT parameters				
Pixel spacing, mm	0.68 (0.68, 0.73)	0.69 (0.68, 0.73)	0.74 (0.69, 0.78)	.002
Slice spacing, mm				<.001
≤2.5	54 (8.9%)	9 (4.8%)	6 (18.8%)	
3	191 (31.4%)	67 (35.8%)	1 (3.1%)	
5	363 (59.7%)	111 (59.4%)	8 (25.0%)	
10	0	0	17 (53.1%)	

Table 1. Patient characteristics across various data sets. Data are presented as median (interquartile range) or count (percentage) unless otherwise specified. ^aFor patients with multiple tumors, the diameter of the largest tumor was measured.

Prognostic performance of FASP, clin and FASP-clin

FASP demonstrated robust predictive performance for RFS across the test sets (Table 2). The time-dependent ROCs were shown in Supplementary Fig. S6. The C-indices were 0.646 (95% CI: 0.566, 0.725) in the internal test set and 0.786 (95% CI: 0.726, 0.846) in the external test set. In contrast, the clinical model based on non-tumor-derived risk factors performed worse, with C-indices of 0.617 (95% CI: 0.542, 0.692) and 0.717 (95% CI: 0.638, 0.795) in the internal and external test sets, respectively (both adjusted $P<.001$). When integrating the FASP score with clinical factors (FASP-Clin), predictive performance further improved, yielding C-indices of 0.664 (95% CI: 0.591, 0.736) in the internal test set and 0.800 (95% CI: 0.750, 0.849) in the external test set (both adjusted $P<.001$).

Additionally, when the tumor masks generated by the automatic tumor segmentation model were replaced with ground-truth annotations by professional radiologists, both FASP and FASP-Clin achieved slightly higher C-indices in both test sets.

Using the median risk score from the development set as the cutoff, FASP effectively stratified patients into low- and high-risk groups in both test sets (Fig. 2).

Univariable and multivariable analyses of the FASP score and clinical factors

In univariable analyses, a higher FASP score was significantly correlated with increased recurrence risk in both test sets (Supplementary Table S4). In multivariable analyses incorporating FASP score and non-tumor-derived clinical risk factors (cirrhosis status, AFP, and ALBI), the FASP score remained an independent prognostic indicator in both the internal test set (hazard ratio [HR]=1.56, [95% CI: 1.19, 2.05], $P=.001$) and the external test set (HR=3.02, [95% CI: 1.62, 5.62], $P<.001$).

Comparison of FASP with tumor-based methods

Overall, FASP outperformed all three tumor-based models (tumorSemantics, tumorRadiomics, tumorSurv) across both test sets (all adjusted $P<.05$, Table 3). The time-dependent ROCs were displayed in Supplementary Fig. S7. The C-indices of FASP were 0.646 (95% CI: 0.566, 0.725) in the internal test set and 0.786 (95% CI: 0.726, 0.846) in the external test set, while the C-indices of the tumor-based models ranged from 0.623 to 0.632 in the internal test set and 0.523 to 0.775 in the external test set.

In a subgroup analysis based on time to recurrence (2 years after surgery as the cutoff), FASP demonstrated predictive capability for both early and late recurrence. For early recurrence, FASP achieved C-indices of 0.664 (95% CI: 0.575, 0.753) and 0.789 (0.724, 0.853) in the internal and external test sets, respectively. FASP was superior to all three tumor-based models in both test sets, except that it was comparable to tumorSemantics in the internal test set. Regarding late recurrence, FASP achieved C-indices of 0.541 (95% CI: 0.401, 0.682) and 0.778 (95% CI: 0.614, 0.942) in the two test sets. FASP surpassed all three tumor-based models in both test sets, except for showing non-inferiority to tumorSemantics in the external test set.

Correlations between FASP score and staging systems and pathological risk factors

Correlation analyses between the FASP score and established staging systems were conducted in the internal test set. The FASP score showed strong correlations with Hong Kong Liver Cancer (HKLC) system ($Rho=0.63$, [95% CI: 0.45, 0.75]), Barcelona Clinic Liver Cancer (BCLC) ($Rho=0.38$, [95% CI: 0.17, 0.56]) and American Joint Committee on Cancer (AJCC) tumor-node-metastasis (TNM) system T stage ($Rho=0.47$, [95% CI: 0.28, 0.63]) (all $P<.001$, Supplementary Table S5). These analyses were not performed for the external test set due to data insufficiency.

Additionally, the FASP score was significantly correlated with pathological risk factors, including tumor differentiation ($Rho=0.21$, [95% CI: 0.07, 0.35], $P=.004$) and micronecrosis score ($Rho=0.36$, [95% CI: 0.23, 0.48], $P<.001$).

Visual explanations of the LiverSurv model

CAMs generated using Grad-CAM provided insights into regions contributing to FASP’s recurrence risk predictions (Fig. 3). These visualizations highlighted both tumor and non-tumor regions, supporting the biological and clinical relevance of the model.

Model	Internal test set				External test set			
	C-index	Delta C-index	P Value	Adjusted P Value ^b	C-index	Delta C-index	P Value	Adjusted P Value ^b
FASP	0.646 (0.566, 0.725)	Ref	Ref	Ref	0.786 (0.726, 0.846)	Ref	Ref	Ref
Clin	0.617 (0.542, 0.692)	−0.036 (−0.046, −0.025) ^c	<.001 ^c	<.001 ^c	0.717 (0.638, 0.795)	−0.072 (−0.086, −0.058) ^c	<.001 ^c	<.001 ^c
FASP-Clin	0.664 (0.591, 0.736)	0.017 (0.015, 0.019) ^c	<.001 ^c	<.001 ^c	0.800 (0.750, 0.849)	0.013 (0.008, 0.019) ^c	<.001 ^c	<.001 ^c
FASP_inferWtGT ^a	0.653 (0.575, 0.731)	0.007 (0.005, 0.009) ^c	<.001 ^c	<.001 ^c	0.795 (0.733, 0.856)	0.009 (0.005, 0.013) ^c	<.001 ^c	<.001 ^c
FASP-Clin_inferWtGT ^a	0.672 (0.601, 0.743)	0.008 (0.005, 0.010) ^d	<.001 ^d	-	0.818 (0.772, 0.864)	0.020 (0.017, 0.022) ^d	<.001 ^d	-

Table 2. Performance measurements of FASP, Clin and FASP-Clin. Data in parentheses are 95% CIs. ^aDuring inference, the tumor masks from the automatic segmentation model were replaced with the ground truth tumor masks provided by professional radiologists. ^bAdjusted P values using Bonferroni correction for 3 comparisons. ^cCompared to FASP. ^dCompared to FASP-Clin.

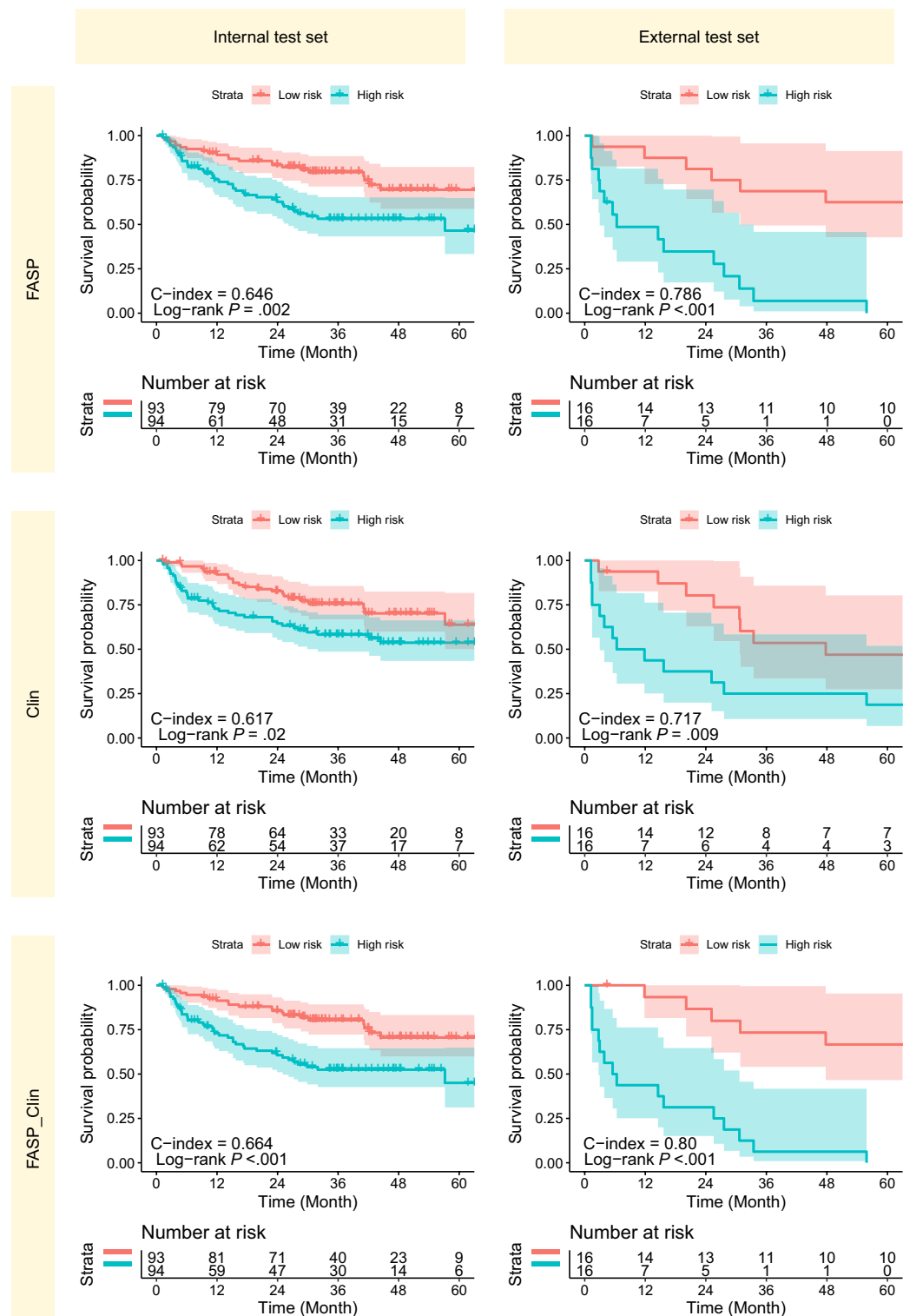


Fig. 2. Kaplan-Meier plots of FASP, Clin and FASP-Clin across the internal and external test sets.

FASP primarily focused on tumor areas, with saliency maps suggesting intra-tumor heterogeneity. Some patients showed higher activation in the central tumor area, while others exhibited heightened activation along tumor periphery (Fig. 3).

Beyond tumor regions, FASP identified clinically significant patterns in non-tumor liver tissue (Fig. 3). In one patient, CAMs emphasized interactions between tumors and peritumoral tissues, with the activation increasing closer to the tumor margin. Additionally, FASP recognized the contribution of smaller tumors that were overlooked by conventional tumor-based methods.

Recurrence type	Internal test set				External test set			
	C-index	Delta C-index	P Value	Adjusted P Value ^a	C-index	Delta C-index	P Value	Adjusted P Value ^a
Overall								
tumorSemantics	0.630 (0.557, 0.704)	−0.017 (−0.023, −0.010)	<.001	<.001	0.775 (0.712, 0.839)	−0.009 (−0.016, −0.002)	0.009	.03
tumorRadiomics	0.632 (0.557, 0.707)	−0.015 (−0.021, −0.009)	<.001	<.001	0.523 (0.406, 0.640)	−0.279 (−0.293, −0.265)	<.001	<.001
tumorSurv	0.623 (0.551, 0.694)	−0.029 (−0.034, −0.024)	<.001	<.001	0.763 (0.692, 0.833)	−0.025 (−0.033, −0.016)	<.001	<.001
FASP	0.646 (0.566, 0.725)	Ref	Ref	Ref	0.786 (0.726, 0.846)	Ref	Ref	Ref
Early recurrence (≤2year)								
tumorSemantics	0.662 (0.581, 0.743)	−0.004 (−0.011, 0.004)	.340	1	0.772 (0.692, 0.852)	−0.012 (−0.019, −0.004)	.002	.006
tumorRadiomics	0.653 (0.570, 0.736)	−0.012 (−0.019, −0.006)	<.001	.001	0.571 (0.414, 0.728)	−0.232 (−0.249, −0.215)	<.001	<.001
tumorSurv	0.645 (0.565, 0.724)	−0.025 (−0.031, −0.019)	<.001	<.001	0.770 (0.683, 0.856)	−0.016 (−0.024, −0.007)	<.001	<.001
FASP	0.664 (0.575, 0.753)	Ref	Ref	Ref	0.789 (0.724, 0.853)	Ref	Ref	Ref
Late recurrence (>2year)								
tumorSemantics	0.447 (0.317, 0.577)	−0.095 (−0.110, −0.080)	<.001	<.001	0.784 (0.660, 0.909)	−0.008 (−0.025, 0.009)	0.358	>.99
tumorRadiomics	0.514 (0.382, 0.646)	−0.030 (−0.045, −0.015)	<.001	<.001	0.393 (0.207, 0.579)	−0.393 (−0.417, −0.370)	<.001	<.001
tumorSurv	0.496 (0.364, 0.629)	−0.049 (−0.060, −0.037)	<.001	<.001	0.744 (0.575, 0.912)	−0.051 (−0.069, −0.033)	<.001	<.001
FASP	0.541 (0.401, 0.682)	Ref	Ref	Ref	0.778 (0.614, 0.942)	Ref	Ref	Ref

Table 3. Comparisons of FASP and tumor-based models. Data in parentheses are 95% CIs. ^aAdjusted *P* values using Bonferroni correction for 3 comparisons.

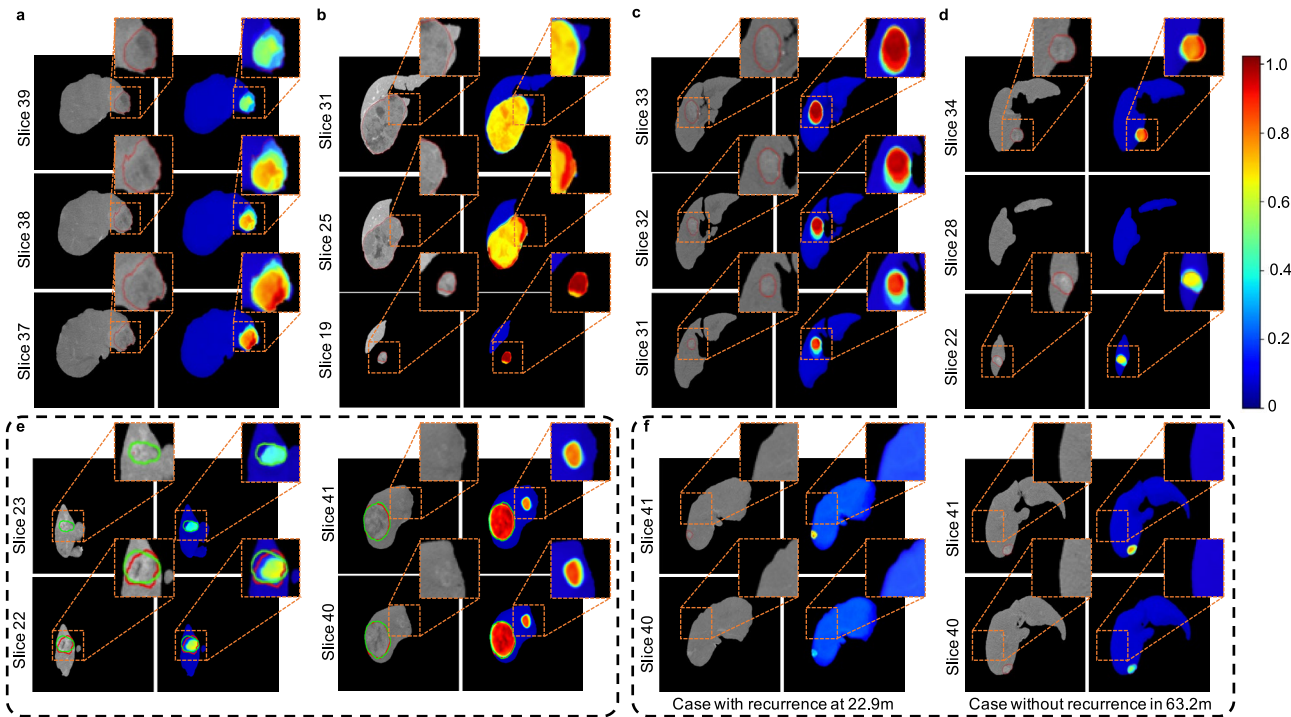


Fig. 3. Class saliency maps of eight patients to show recurrence-related characteristics identified by FASP. The color gradient from blue to red indicates increasing risk. (a) Intra-tumor heterogeneity with the highest activation localized at the tumor center. (b) Intra-tumor heterogeneity with the highest activation at the tumor margin. (c) Peritumoral area contributing to recurrence. (d) Multifocality contributing to recurrence. (e) **Left:** FASP identified the entire tumor area though some slices were missed by the tumor segmentation model (Red). **Right:** FASP detected an additional tumor that was ignored by both the tumor segmentation model (Red) and human readers (Green). (f) Two patients with similar clinical features but divergent outcomes, possibly due to different underlying liver disease patterns as identified by FASP.

Notably, FASP demonstrated resilience to errors in tumor segmentation (Fig. 3). In one patient, it correctly identified a top slice of a tumor missed by the automatic segmentation model. In another patient, FASP uncovered a hidden tumor overlooked by both human readers and segmentation algorithms.

Additionally, FASP showed the ability to account for the underlying liver disease patterns (Fig. 3). For example, in two male patients with cirrhosis and small tumors (<3 cm) but divergent outcomes, the clinical model incorrectly assigned a higher risk to the patient with better RFS. In contrast, FASP correctly assigned a higher risk to the patient with worse RFS, consistent with pathological findings of severe liver diseases, including diffuse cirrhosis and liver atrophy.

Discussion

In addition to tumor-specific features, background liver conditions, such as cirrhosis, play a significant role in HCC recurrence. However, previous imaging-based studies have primarily focused on the largest tumor for RFS prediction. In this multicenter study, we developed and validated FASP, an AI-driven whole-liver imaging modeling system designed to automatically predict recurrence risk in primary HCC from preoperative CECT images. FASP obtained C-indices of 0.646 and 0.786 in the internal and external test sets, respectively, outperforming three tumor-based models built using conventional strategies.

Whole-liver imaging for survival prediction has rarely been explored. Xu et al. attempted to analyze 2D liver ROIs for early recurrence prediction using a ResNet-based model, but their approach could not be applied to the 3D whole-liver volume²³. Studies in the lung, pancreas, and brain have demonstrated that whole-organ analytics can offer comprehensive insights into disease biology^{22,24–26}. The unique challenges of whole-liver analysis arise from the liver's large size, variable shape, complex anatomy and diverse pathologies^{11,27}. This study developed FASP to successfully extract whole-liver information from CECT to enhance HCC recurrence prediction, demonstrating the feasibility and utility of whole-liver imaging assessment.

FASP offers several advantages over conventional tumor-based approaches. First, it captures recurrence-related information from the entire liver, outperforming all three tumor-based models. Second, FASP is fully automated and eliminates the need for labor-intensive human annotation. Third, since FASP does not rely on manual annotation, it holds the potential to be applicable to tumor types with unclear boundaries. Finally, using whole-liver ROIs as input, FASP preserves the original shape and texture of both liver and tumor regions, enabling whole-liver interpretability and enhancing clinical applicability.

Our findings suggest that assessing cirrhosis severity as a continuous variable may be more informative for predicting HCC recurrence. We found no evidence of a significant association between binary cirrhosis status and recurrence risk, consistent with prior studies^{18,28,29}. This might be because cirrhosis is actually a progressive disease with varying degrees of severity rather than a strict binary condition²⁸. It has been reported that histological subclassification of cirrhosis can further stratify HCC patients into distinct recurrence risk groups²⁹. The superior performance of FASP over the clinical model may be attributed to its ability to capture whole-liver characteristics, potentially reflecting the overall severity of cirrhosis.

The FASP score was significantly correlated with poor tumor differentiation and higher micronecrosis score. Poorly differentiated HCC tumors are known to be more aggressive and prone to recurrence³⁰. Tumor micronecrosis, a microscopic feature reflecting chronic ischemic injury and angiogenesis, is linked to poor RFS³¹. Our findings suggest that FASP captures imaging features indicative of tumor aggressiveness and the pathological microenvironment.

This study has several limitations. First, although a substantial dataset was used for model training and internal validation, the external test set was relatively small and acquired with larger slice thickness. This might reduce the statistical power of the estimated model performance metrics, and therefore the external validation results should be interpreted with caution. Expanding the external dataset could further evaluate the model's robustness and generalizability. Second, while FASP demonstrated promising predictive performance, its accuracy could be improved through strategies such as incorporating multimodal data, such as multi-parameter magnetic resonance imaging (MRI). Third, although Grad-CAM was applied to visualize the decision-making process of FASP, a method offering clearer, quantitative interpretability is needed in future work.

In conclusion, this study presented FASP, a fully automated prognostic system that leverages whole-liver imaging information from preoperative CECT to predict recurrence risk after curative resection in HCC patients. By integrating CECT features from both tumor regions and the non-tumoral liver environment, FASP outperformed tumor-based models in both internal and external test sets, emphasizing the importance of comprehensive liver assessment in predicting HCC recurrence. Future research could focus on further improving whole-liver modeling performance and enhancing model interpretability.

Data availability

Data from the TCGA-LIHC data collection is publicly available at <https://www.cancerimagingarchive.net/collection/tcga-lihc/>. Requests to access other datasets used in this study for research purposes should be directed to the corresponding author (email: ljs@zju.edu.cn).

Code availability

The code used in this study is publicly available on GitHub (<https://github.com/huangmozhi/FASP>).

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Author contributions

C.H., Y.T. and J.S.L. conceptualized and designed the study. C.H. conducted the data preprocessing, modeling and statistical analyses with help of Y.T., Y.W.G., P.J.H., Y.Y.X., J.L., W.B.W., and T.S.Z. Q.Z. and Y.Y.W. contrib-

uted to data collection and clinical expertise. M.Z. and J.S.L. jointly supervised the research. C.H., Y.T. and Q.Z. drafted the manuscript. All authors contributed to the experimental design, interpretation of the results, and manuscript revisions. All authors had full access to all the data, and C.H., Y.W.G., Y.Y.W., and J.S.L. directly accessed and verified the data. All authors approved the final manuscript and take full responsibility for its submission.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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